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Steady and dynamic hydroelastic behavior of composite lifting surfaces

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ABSTRACT

We compare the steady-state and dynamic response of composite lifting surfaces in air and in water. Steady results predict lower (resp. higher) load coefficients when the fibers are oriented toward the leading (resp. trailing) edge, as nose-down (resp. nose-up) twist decreases (resp. increases) the effective angle-of-attack, both in air and in water. Natural frequencies are lower, and damping coefficients are higher, in water compared to in air. The natural frequencies and damping coefficients vary with speed. The speed and frequency dependency of the fluid loads lead to an emergence of a new mode. The vibration frequency of this new mode could be near zero and much significantly than the fundamental system frequency in quiescent water. Static divergence occurs when the frequency and damping value of the new mode goes to zero, while the frequencies of the original structural modes are nonzero. Unlike conventional coupled-mode flutter that involves frequency coalescence of two adjacent modes, composite plates in water tend to experience single-mode flutter, when the damping of the new low-frequency mode vanishes. In general, static-divergence governs when material bend-twist coupling leads to nose-up twist, and flutter governs otherwise. Dynamic load amplification is more severe in water because of higher fluid added mass.

1. Introduction

Most marine and aerospace systems use lift generating surfaces to maneuver, propel, cruise, and harvest energy. These lift generating surfaces need to be strong and stable enough to withstand the fluid loads. When used as propulsion and flow control devices, it is critical to minimize flow-induced vibrations and dynamic load amplifications, whereas flow-induced vibrations are typically desired for energy harvesting applications.

Flow-induced forces depend on the fluid density and the relative flow speed. Since the water density is about 850 times higher than that of air, the mechanical design of marine structures can be more stringent than in air because of the higher hydrodynamic loads. It is for this reason that:

1) Most lifting surfaces used in marine systems have thicker cross sections and smaller span-to-chord ratios than their counterparts in the aerospace or wind-energy harvesting systems. Similarly, the high fluid loading in water typically requires solid cross sections instead of hollow or foam-core cross sections that are more common for aerospace structures,
2) Use of powerful hydraulic actuators to actively deflect or control these lifting surfaces in water are limited to low actuation frequencies (typically in the order of 1 Hz or less), because of high power required associated with the high fluid loads, and
3) There exist additional operation and maintenance concerns related to hydraulic leaks and accelerated fatigue of various component systems.

Cavitation can develop in water when the local pressure drops to near the saturated vapor pressure of the fluid, which can induce highly unsteady and nonlinear loads, noise, vibration, and erosion on the lifting surfaces during high speed operating conditions near the sea level (refs. [1,2]). Cavitation can also occur in the small gaps between marine structures (e.g. flaps or junction areas), where the local flow may attain high speeds. Ventilation, when gas is drawn to low pressure regions around the body, can also occur for high speed operations near the water surface, which can lead to sudden and large drops in lift or side force, as well as flow-induced instabilities (ref. [3,4]). In marine environments, random sea states can induce stochastic load variations [5]. Due to all these complications, it is critical to consider both the steady-state and dynamic hydroelastic response when designing marine lifting surfaces.

Typically, marine structures are made of engineered materials such

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as metallic alloys, which are relatively stiff, but heavy and prone to corrosion in seawater. Alternatively, it is possible to use composite materials, such as carbon fiber reinforced polymers (CFRP) and glass fiber reinforced polymers (GFRP), which can carry very high loads in their strong fiber directions. Composite materials are often much lighter, but they can have the same or higher strength than metal alloys. Composites also show superior long-term resistance to corrosion and higher fatigue resistance compared to metal alloys (ref. [6]). The disadvantage of the composite structures is that their large-scale manufacturing process is highly customized compared to metal structures, which makes them expensive and takes longer to build; these challenges are progressively being eased with advances in automatic composite layering robotic technologies. Hence, there has been an emerging interest to design marine lifting surfaces with composite materials. Relevant research has shown that adaptive composite lifting surfaces can deliver superior performances compared to their rigid counterparts in off-design conditions or in spatially or temporally varying inflows, as they may passively morph their instantaneous flying shapes to adapt to changes in the flow conditions without the use of bulky and heavy hydraulic actuators that are limited to low actuation frequencies (refs. [7–16]). However, for marine structures, most of the past work only focused on analyzing their steady-state performance.

The objective of this paper is to examine both the steady-state and dynamic hydroelastic response and stability of a composite lifting surface in the form of a canonical plate, and to examine the differences between in air and in water operations. Specifically, the results of interest include static and dynamic deformations, load coefficients, natural frequencies, damping loss factors, and the critical speed limits for the onset of flutter and static-divergence instabilities.

The physical, mathematical, and numerical formulations are given in Section 2; Section 3 shows the validation studies; and Section 4 compares the steady-state and dynamic hydroelastic and aeroelastic results. The conclusions and future work are stated in Section 5.

2. Formulation and solution algorithm

2.1. Beam equations with bend-twist coupling and warping effects

To focus on the basic physics, we conduct our studies on a simplified cantilevered plate geometry, which straightly extends from its clamped edge. The cantilevered plate is meant as a canonical proxy to more complex lifting bodies such as propeller or turbine blades, or other flow control or energy harvesting devices. The plate is assumed to have zero sweep and twist. In a later paper, we will investigate the interplay between sweep and fiber angles.

The plate is assumed to be made of a composite material, which has strong and unidirectional fibers embedded within a weaker matrix material. In general, the composite material may have multiple laminae; for each lamina, the fibers may have different alignments within the plane of the plate. For a multilayered composite, it is possible to find a single effective unidirectional fiber orientation that represents the load-dependent deformation characteristic of the multilayered model (ref. [17]).

We consider the time, t, varying spanwise bending and twisting deformations of the plate, denoted with $h(y,t)$ and $\psi(y,t)$, respectively and shown in Fig. 2.1. These deformations are defined at the elastic axis (E.A.) (i.e. shear center) of the plate. The spanwise y coordinate is zero at the clamped edge; the x coordinate is along the chord (width) of the plate and is zero at the midchord and is parallel to the external flow with speed U_0; the z coordinate is along the thickness of the plate and satisfies $\vec{z} = \vec{x} \times \vec{y}$. Therefore, h is positive along the z-axis, and ψ is positive along the y-axis. The span length of the plate is L, the thickness of the plate is q, and the chord length (width) of the plate is $2b$.

The coupled differential equations governing the beam spanwise bending and twisting deformations are given in Eqs. (1) and (2):

$$m_s\frac{\partial^2 h}{\partial t^2} - m_s x_\alpha b\frac{\partial^2 \psi}{\partial t^2} = -EI_s\frac{\partial^4 h}{\partial y^4} - K_s\frac{\partial^3 \psi}{\partial y^3} - abEI_s\frac{\partial^4 \psi}{\partial y^4} + F_z^{fluid} + F_z^{exc}, \tag{1}$$

$$I_s^{E.A.}\frac{\partial^2 \psi}{\partial t^2} - m_s x_\alpha b\frac{\partial^2 h}{\partial t^2} = +GJ_s\frac{\partial^2 \psi}{\partial y^2} + K_s\frac{\partial^3 h}{\partial y^3} - abEI_s\frac{\partial^4 h}{\partial y^4} - S_s\frac{\partial^4 \psi}{\partial y^4} + M_y^{fluid} + M_y^{exc}, \tag{2}$$

where m_s is the mass of the plate per unit span length; $I_s^{E.A.}$ is the mass moment inertia of the plate along the E.A. per unit span length; EI_s and GJ_s are, respectively, the sectional bending and torsion stiffness of the beam. In Eqs. (1) and (2):, K_s is the bend-twist coupling stiffness, which is nonzero if the fibers are laid at non-zero angle θ from the spanwise axis; θ is defined as positive towards the plate leading edge with respect to the incoming flow with speed U_0 along the x direction, as shown in Fig. 2.1. S_s in Eq. (2) is the warping resistance of the rectangular beam, which is $S_s = EI_s b^2\left(\frac{1}{3} + a^2\right)$ (ref. [9]). For composite plates, stiffness parameters EI_s, GJ_s, and K_s depend not only on the plate geometry, but also on the orientation of the fibers, as well as the number and arrangement of the laminae; these relations are given in the Appendix A and in ref. [17]. Ref. [9] has shown that for a cantilever plate representing a typical airfoil with a span to chord ratio $L/2b = 20/3$, warping effects can noticeably change the predicted stability boundaries. Compared to airfoils, typical hydrofoils have a much lower span to chord ratio because the much higher loading in water, as water density is about 850 times higher than air. Hence, results are shown in the next section are for shorter plates with $L/2b = 3$, which is more typical of hydrofoils. Note that with the smaller $L/2b$, warping effects tend to be more important, as the warping terms scale with $1/L^4$ according to Eqs. (1) and (2). In addition, as shown in Appendix B, the predicted natural frequencies and the critical stability boundaries are higher when warping effects are considered compared to classical beam model results without warping. It should also be noted that while composite plate theory would be more appropriate for low-aspect ratio marine propeller blades, composite beam theory is deemed sufficient to predict the lower-order modes and vibration characteristics of marine control surfaces and energy harvesting devices with mean span-to-chord ratio greater than or equal to three.

In Eqs. (1) and (2), $x_\alpha b$ is the distance to the sectional center of gravity (C.G.) from the E.A., and $x_\alpha b$ is positive if the centroid is aft of the E.A. Similarly, the length ab is the distance from the midchord to the E.A., and ab is positive if E.A. is aft of the midchord. Both $x_\alpha b$ and ab are also shown in the front view given in Fig. 2.2 with the flow going from right to left. Finally, in Eqs. (1) and (2), $F_z^{fluid}(y,t)$ and $M_y^{fluid}(y,t)$ are the sectional flow-induced force and moment terms (respectively along the z and y directions), caused by uniform external flow with speed U_0. We apply known excitation forces and moments F_z^{exc} and M_y^{exc} to the right-hand-side of the equation of motion in Eqs. (1) and (2).

The boundary conditions for this clamped-free plate geometry are mathematically expressed as:

$$h|_{y=0} = 0,\ \psi|_{y=0} = 0,\ \left.\frac{\partial h}{\partial y}\right|_{y=0} = 0,\ \left.\frac{\partial \psi}{\partial y}\right|_{y=0} = 0, \tag{3}$$

$$EI_s\left.\frac{\partial^3 h}{\partial y^3}\right|_{y=L} + K_s\left.\frac{\partial^2 \psi}{\partial y^2}\right|_{y=L} + abEI_s\left.\frac{\partial^3 \psi}{\partial y^3}\right|_{y=L} = 0, \tag{4}$$

$$EI_s\left.\frac{\partial^2 h}{\partial y^2}\right|_{y=L} + K_s\left.\frac{\partial \psi}{\partial y}\right|_{y=L} + abEI_s\left.\frac{\partial^2 \psi}{\partial y^2}\right|_{y=L} = 0, \tag{5}$$

$$GJ_s\left.\frac{\partial \psi}{\partial y}\right|_{y=L} + K_s\left.\frac{\partial^2 h}{\partial y^2}\right|_{y=L} - abEI_s\left.\frac{\partial^3 h}{\partial y^3}\right|_{y=L} - S_s\left.\frac{\partial^3 \psi}{\partial y^3}\right|_{y=L} = 0, \tag{6}$$

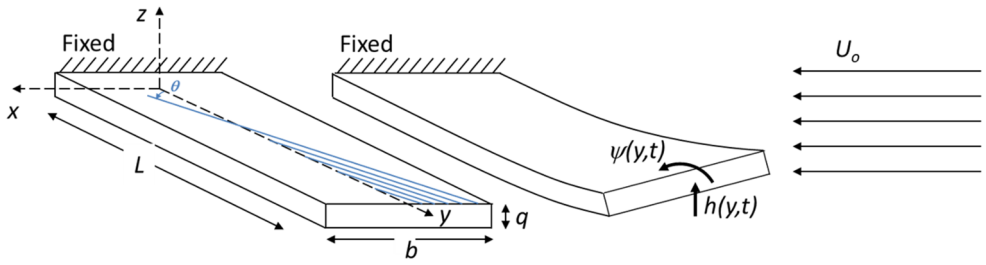


Fig. 2.1. A straight cantilever beam embedded with unidirectional fibers aligned at an angle θ from the spanwise axis, and is undergoing spanwise bend and twist deformations, $h(y,t)$ and $\psi(y,t)$, respectively, due to uniform external flow with speed U_o.

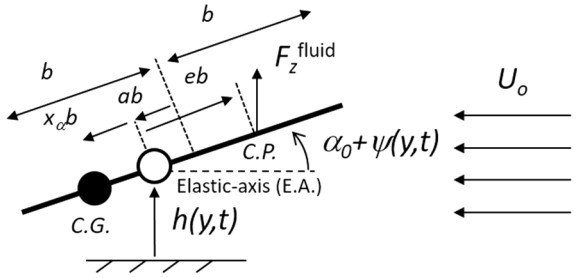


Fig. 2.2. Front 2D view of a cross-section of the plate shown in Fig. 2.1 with spanwise bend and twist deformations, h and ψ, defined; elastic axis (E.A.); center of gravity (C.G.); center of pressure (C.P.), which is assumed to be at $b/2$ downstream from the plate leading edge.

$$S_s \frac{\partial^2 \psi}{\partial y^2}\bigg|_{y=L} + abEI_s \frac{\partial^2 h}{\partial y^2}\bigg|_{y=L} + abK_s \frac{\partial \psi}{\partial y}\bigg|_{y=L} = 0 \tag{7}$$

These boundary conditions and equations of motion (Eqs. (1) and (2)) are derived by minimizing the virtual work of the composite beam's strain energy [9]. The boundary conditions in Eq. (3) requires the deformations and slopes of deformations to be zero at the clamped edge. Note that classical formulations for the beam torsion do not specify zero-slope for the twist angle at the clamped edge [18]. This additional boundary condition arises due to the increased order of the differential equation with the warping effect and the effect of having a shear center away from the midchord. We simplify Eqs. (1) and (2) and the fluid force models presented next, by assuming a separation of variables for h and ψ in time and in space:

$$h(y,t) = H(y,\omega)\exp(i\omega t), \text{ and}$$
$$\psi(y,t) = \Psi(y,\omega)\exp(i\omega t) \tag{8}$$

where ω is the complex frequency of the deformations (similar to Laplace variable). If ω is purely real, the deformations would have a constant amplitude and oscillate at a frequency ω; if ω has a positive imaginary component, the deformations would decay with a frequency $\mathrm{Re}(\omega)$, and with a damping loss factor η:

$$\eta = 2\,\mathrm{Im}(\omega)/\mathrm{Re}(\omega). \tag{9}$$

Flutter instability occurs for $\eta \le 0$, i.e., $\mathrm{Im}(\omega) \le 0$ with $\mathrm{Re}(\omega) > 0$, so that the oscillations increase in time. Static-divergence instability occurs for $\mathrm{Re}(\omega) = 0$, $\mathrm{Im}(\omega) \le 0$, i.e., when the total effective stiffness of the plate is less than or equal to zero, or when the fluid disturbing moment is equal to or exceeds the elastic restoring moment (ref. [19]).

2.2. Unsteady potential flow model

The fluid model uses an incompressible and inviscid potential flow formulation based on Yates' work in ref. [20], which is the three-dimensional (3D) extension of Theodorsen's method derived in refs. [21] and [22]. The original Theodorsen's method is a two-dimensional (2D) potential flow model of the unsteady fluid forces, which assumes that the plate deformations are small and harmonic, and the flow is fully attached without any flow separation. For the fluid flow calculations, the plate is assumed to be symmetric and infinitely thin, the center of pressure is at quarter chord downstream from the foil leading edge, and the shed vortices convect downstream parallel to the inflow direction. The resultant fluid lift and moment expressions are given in Eqs. (10) and (11):

$$\begin{aligned} F_z^{fluid} &= -\pi\rho_f b^2 \frac{\partial^2 h}{\partial t^2} - \pi\rho_f b^2 ab \frac{\partial^2 \psi}{\partial t^2} \\ &\quad - \tfrac{1}{2}\rho_f U_0 a_0 2b\Omega(k)\frac{\partial h}{\partial t} + \tfrac{1}{2}\rho_f U_0 b^2 [2\pi + a_0[1-2a]\Omega(k)]\frac{\partial \psi}{\partial t} \\ &\quad + \tfrac{1}{2}\rho_f U_0^2 a_0 2b\Omega(k)[\psi + \alpha_0], \end{aligned} \tag{10}$$

$$\begin{aligned} M_y^{fluid} &= -\pi\rho_f b^2 ab \frac{\partial^2 h}{\partial t^2} - \pi\rho_f b^4 \left[\tfrac{1}{8} + a^2\right]\frac{\partial^2 \psi}{\partial t^2} \\ &\quad - \tfrac{1}{2}\rho_f U_0 a_0 [eb] 2b\Omega(k)\frac{\partial h}{\partial t} \\ &\quad - \tfrac{1}{2}\rho_f U_0 \tfrac{b^2}{2}[1-2a][2\pi b - a_0 2eb\Omega(k)]\frac{\partial \psi}{\partial t} \\ &\quad + \tfrac{1}{2}\rho_f U_0^2 a_0 [eb] 2b\Omega(k)[\psi + \alpha_0]. \end{aligned} \tag{11}$$

for $h(y,t) = H(y,\omega)\exp(i\omega t)$ and $\psi(y,t) = \Psi(y,\omega)\exp(i\omega t)$. In Eqs. (10) and (11):

$$k = \frac{\omega b}{U_0}; \quad \Omega(k) = \frac{H_1^2(k)}{H_1^2(k) + iH_0^2(k)} \tag{12}$$

ρ_f is the fluid density and $\Omega(k)$ is the Theodorsen function, which consists of Bessel functions of the third kind (H_0^2 and H_1^2). $\Omega(k)$ appears in Eqs. (10) and (11) within the circulatory terms to model the effect of the shed vorticities in the wake on the plate (and they are responsible for causing nonlinear memory or history-dependent effects). $\Omega(k)$ has complex values; it has a positive real component that ranges from $\Omega = 1$ for $k = 0$ to $\Omega = 1/2$ for $\mathrm{Re}(k) \to \infty$; in general, k itself can take complex values following the note in ref. [23].

Note in Eqs. (10) and (11) that the lift and moment have non-circulatory terms as well. a_0 is the initial geometric angle of attack of the plate with respect to the inflow U_0; a_0 is the non-dimensional slope of the steady lift coefficient (where the sectional lift is scaled with $\frac{1}{2}\rho_f U_0^2 2b$) versus a_0 curve. a_0 is taken as 2π in Theodorsen's formulation in 2D, but has a value lower than 2π for 3D wings with finite-span due to downwash effects caused by the free vortices in the wake. In addition, in 3D, a_0 is non-uniform along the span. The length eb defined in Eq. (13) is the distance from the elastic axis (E.A.) to the center-of-pressure (C.P.), where the resultant lift acts, as shown in Fig. 2.2, and eb is positive when pointing towards the leading edge; for our case, the C.P. is assumed to be at the quarter-chord distance from the plate leading edge, i.e.:

$$eb = \left(a + \frac{1}{2}\right)b \tag{13}$$

Note that eb can take different values for cambered foils.

The fluid force/moment terms in Eqs. (10) and (11) are combinations of the 1st and 2nd time-derivatives of the deformations, and the deformations themselves. The terms in phase with the foil acceleration

and velocity are respectively called the fluid added-mass and potential flow-induced damping (as viscous effects are ignored in this work) forces. The total effective mass of the plate increases due to the fluid-added mass effect, particularly for lifting surfaces in water. Since $\Omega(k)$ has positive real components, the fluid de-stiffening force terms, $\frac{1}{2}\rho_f U_0^2 a_0 2b\Omega(k)(\psi + \alpha_0)$ and $\frac{1}{2}\rho_f U_0^2 a_0 eb2b\Omega(k)(\psi + \alpha_0)$ in Eqs. (10) and (11), respectively, has a de-stiffening effect when carried over to the left hand side of Eqs. (1) and (2), so they act to increase the foil deformations. Eqs. (10) and (11) show that, in addition to bend-twist coupling due to material anisotropy, the presence of fluid-induced loads induce additional hydrodynamic bend-twist coupling. The fluid added mass term does not depend on U_0, but the fluid damping terms are proportional to $\rho_f U_0$, while the fluid de-stiffening terms are proportional to $\rho_f {U_0}^2$. The dependence of all the fluid terms to ρ_f implies that the fluid forces are relatively much more important in water than in air, particularly for lightweight structures. For lifting surfaces in water, the growth of the fluid damping and de-stiffening terms with U_0 are more significant than in air because of the dependence on ρ_f.

To extend Theodorsen's 2D method to 3D following Yates (ref. [20]), the spanwise variation in the lift slope, a_0, for the circulatory terms in Eqs. (10) and (11) is calculated using the steady lifting-line formulation for finite-span wings from Glauert in Ref. [24]. The formulation first assumes a symmetric circulation distribution about the clamped edge of the plate with a finite span. The spanwise circulation distribution, Γ, is expanded in Fourier series as:

$$\Gamma = 4U_0 L \sum_{n=1,3,5,\ldots}^{\infty} A_n \sin(n\phi), \tag{14}$$

where $\phi \in [0, \pi]$ is defined as:

$$y = -L\cos(\phi). \tag{15}$$

for a symmetric double body (across the clamped edge). The constants A_n's in Eq. (14) are obtained by solving Eq. (16), which accounts for the downwash effect caused by the trailing vortex sheet. Once the A_n's are found, the spanwise lift slope (a_0) in Eqs. (10) and (11) for a finite-span plate is predicted using Eq. (17):

$$\frac{\pi b}{2L}\alpha_0 \sin(\phi) = \sum_{n=1,3,5\ldots}^{\infty} A_n \sin(n\phi)\left\{\frac{\pi bn}{2L} + \sin(\phi)\right\} \tag{16}$$

$$a_0 = \frac{F_{y0}^{fluid}}{\frac{1}{2}\rho_f U_0^2 2b\alpha_0} = \frac{\rho_f U_0 \Gamma}{\frac{1}{2}\rho_f U_0^2 2b\alpha_0} = \frac{\Gamma}{U_0 b\alpha_0} \tag{17}$$

2.3. Solution algorithm

Eqs. (1) and (2) with Eqs. (10) and (11) are solved in the frequency domain, by using the decomposition noted in Eq. (8), which simplifies the time-derivatives as:

$$\frac{\partial h}{\partial t} = i\omega h, \quad \frac{\partial^2 h}{\partial t^2} = -\omega^2 h, \quad \frac{\partial \psi}{\partial t} = i\omega\psi, \quad \frac{\partial^2 \psi}{\partial t^2} = -\omega^2\psi \tag{18}$$

The coupled plate Eqs. (1) and (2) are now two 4th order ordinary differential equations in the spanwise coordinate y. We transform these equations to 8 first-order differential equations in state space. The numerical procedure to solve this eigenvalue problem involved solving the ordinary differential equations in the y coordinate with the boundary conditions (Eqs. (3)–(7)), while ignoring the initial angle of attack, α_0. We use MATLAB®'s ode45 function to solve the equations after combining it with the specific shooting method described in [25] together with the Secant method.

The steady-state deformations and loads due to non-zero α_0 are obtained by taking $\Omega(k) = 1$ and removing the time derivative terms, yielding Eqs. (19)–(22):

$$0 = -EI_s\frac{\partial^4 h}{\partial y^4} - K_s\frac{\partial^3 \psi}{\partial y^3} - abEI_s\frac{\partial^4 \psi}{\partial y^4} + F_{z0}^{fluid}, \tag{19}$$

$$0 = GJ_s\frac{\partial^2 \psi}{\partial y^2} + K_s\frac{\partial^3 h}{\partial y^3} - abEI_s\frac{\partial^4 h}{\partial y^4} - S_s\frac{\partial^4 \psi}{\partial y^4} + M_{y0}^{fluid} \tag{20}$$

with,

$$F_{z0}^{fluid} = \tfrac{1}{2}\rho_f U_0^2 a_0 2b(\psi + \alpha_0), \tag{21}$$

$$M_{y0}^{fluid} = \tfrac{1}{2}\rho_f U_0^2 a_0 eb2b(\psi + \alpha_0). \tag{22}$$

In Eqs. (19)–(22)), ${F_{z0}}^{fluid}$ and ${M_{y0}}^{fluid}$ are the steady-state fluid lift force and moments. We define the nondimensional steady lift and moment coefficients in 3D as:

$$C_{L0} = \frac{\int_0^L F_{z0}^{fluid} dy}{\frac{1}{2}\rho_f U_0^2 (2b)L}, \quad C_{M0} = \frac{\int_0^L M_{y0}^{fluid} dy}{\frac{1}{2}\rho_f U_0^2 (2b)^2 L}, \tag{23}$$

Note that the subscript 0 is used to denote steady-state results. Eqs. (19)–(23) are solved with the boundary conditions given in Eqs. (3)–(7).

We also predict time-domain results by first computing the transfer function of the coupled system in the Laplace domain subject to initial conditions for bending and twisting deformations, and then by replacing the $i\omega$ terms with Laplace variable s, and the derivative terms with appropriate corrections, e.g.

$$-\omega^2 h \rightarrow s^2 h(s) - sh(t)|_{t=0} - dh(t)/dt|_{t=0}$$

We then predict the time response by computing the numerical inverse Laplace function of the transfer function using the numerical method described in [26].

Finally, we separately compute the frequency spectra of the system. We first excite the system with a constant amplitude external harmonic force and moment with a prescribed real valued frequency ω. We then use the amplitudes of these external forces as ${F_{z0}}^{exc}$ and ${M_{y0}}^{exc}$ to solve Eqs. (1),(2) with the boundary conditions Eqs. (3)–(7) in frequency domain, and compute the amplitudes of the resultant deformations ($H(y,\omega)$ for bending and $\psi(y,\omega)$ for twisting). The differential equations in frequency domain are given in Eqs. (24),(25).

$$-\omega^2 m_s H + \omega^2 m_s x_\alpha b\,\psi = -EI_s\frac{\partial^4 H}{\partial y^4} - K_s\frac{\partial^3 \psi}{\partial y^3} - abEI_s\frac{\partial^4 \psi}{\partial y^4} + F_z^{fluid} + F_{z0}^{exc}, \tag{24}$$

$$-\omega^2 I_s^{E.A.}\psi + \omega^2 m_s x_\alpha bH = GJ_s\frac{\partial^2 \psi}{\partial y^2} + K_s\frac{\partial^3 H}{\partial y^3} - abEI_s\frac{\partial^4 H}{\partial y^4} - S_s\frac{\partial^4 \psi}{\partial y^4} + M_y^{fluid} + M_{y0}^{exc} \tag{25}$$

where

$$H = H(y,\omega), \quad \Psi = \Psi(y,\omega), \quad F_z^{fluid} = F_z^{fluid}(y,\omega), \quad M_y^{fluid} = M_y^{fluid}(y,\omega)$$

Once we compute $H/{F_{z0}}^{exc}$ and $\psi/{M_{z0}}^{exc}$, we back calculate the flow-induced loads (${F_z}^{fluid}/{F_{z0}}^{exc}$ and ${M_y}^{fluid}/{M_{z0}}^{exc}$) based on the predicted H and ψ.

3. Validations

We validated the current numerical method by comparing its predictions against other relevant published results in the literature.

3.1. Root locus, flutter and static-divergence predictions for an isotropic airfoil in external flow

We validated the flutter and static-divergence speed predictions of our model with the results given in [23] for a 2D isotropic airfoil in

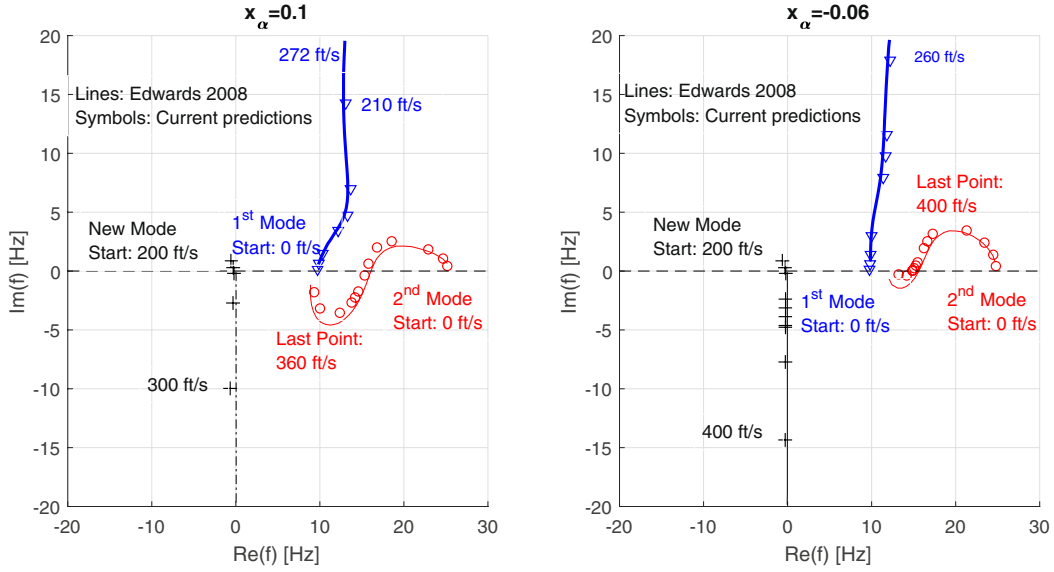


Fig. 3.1. Root locus plots for the complex valued eigenvalues (poles) of the two-dimensional airfoil problem with varying inflow speed U_0 described in ref. [23]; plot on the left is for the case with $x_\alpha = 0.1$ and plot on the right is for the case with $x_\alpha = -0.06$. The real values of the poles, which are the oscillation frequencies, are on the horizontal axes; and the imaginary values of the poles are related to the total damping. In this problem, $2b = 6$ ft, $a = -0.2$, radius of gyration is $0.25b^2$; the in vacuo bending and twisting natural frequencies are $\omega_h = 10$ rad/s, $\omega_\psi = 25$ rad/s; and the material damping is 1.5%. The solid to fluid mass ratio is $m_s/(\rho_f \pi b^2) = 20$.

external flow with bending and twisting degrees of freedom (Ref. [23] further validated their results with several prior studies mentioned therein). The chord length, $2b$, was 6 ft (~1.8 m); the radius of gyration was $0.25b^2$; and $a = -0.2$. For one case, we had $x_\alpha = 0.1$; for another case, we had $x_\alpha = -0.06$. The solid to fluid mass ratio is $\mu = m_s/(\rho_f \pi b^2) = 20$. The material damping for both bending and twisting motions are 1.5%. To be consistent with the results in ref. [23], in this problem, we used the classical beam equations by omitting the S_s and ab terms in the structural stiffness forces in Eqs. (1) and (2) (however, the term a was still included in the fluid loads of Eqs. (10) and (11)). The in vacuo natural frequencies for the first bending and first twisting motions are, respectively, $\omega_h = 10$ rad/s and $\omega_\psi = 25$ rad/s. The plots in Fig. 3.1 show the root locus of the eigenvalues of the system for different flow speeds. At low flow speeds, the damping of the modes increase with speed, as indicated by the increasing imaginary values of the poles. However, for the second mode (primarily torsion mode), at ~150 ft/s (46 m/s) for both cases shown in Fig. 3.1, the damping starts to decrease with further increases in speed and eventually becomes zero, which corresponds to the onset of coupled-mode flutter. In the literature, coupled-mode flutter is also called torsion flutter, and it is characterized by the converging of the frequencies (the real values of the roots) for adjacent bending and torsion modes, where typically the damping of the torsion mode goes towards zero while the damping of the bending mode increases with increasing speed. At $U_0 = 200$ ft/s (61 m/s), a new pole with near zero frequency (close to the positive imaginary axis) appears, and it moves along the negative imaginary axis with further increases in speed; the speed at which the new mode crosses the origin is the static-divergence speed; we will discuss this new mode in more detail later in this paper. It is important to notice that static-divergence may happen, even when the frequencies of the structural modes are not zero. Note that in this work, the term structural modes refer to the modes that are observed in vacuum with zero flow speed. There is a good agreement between our predictions and those presented in [23] for all the modes for both cases. The predicted flutter speeds are: 175 ft/s (53.3 m/s) for the case with $x_\alpha = 0.1$, and 281 ft/s (85.6 m/s) for the case with $x_\alpha = -0.06$; both predicted values are very close to the reported flutter speeds of 169 ft/s (51.5 m/s) and 257 ft/s (78.3 m/s), respectively, in [23]. The static-divergence speeds are similar as well; we predicted 215 ft/s (65.5 m/s), while ref. [23] predicted 217 ft/s (66.1 m/s).

3.2. Flutter and static-divergence speeds of an anisotropic composite wing in external air flow

The second validation example involved comparing our predictions against flutter and static-divergence speeds reported in [9] for an anisotropic composite wing in external air flow. The wing is 20 ft (6.1 m) long, 3 ft (0.91) wide with $a = -0.34$, $EI_s = 23.6 \times 10^6$ lb.ft^2 (9.76 MPa), $GJ_s = 2.39 \times 10^6$ lb.ft^2 (0.988 MPa), $m_s = 0.746$ slug/ft (35.7 kg/m), $I_s^{E.A.} = 1.943$ slug.ft (8.64 kg.m), $x_\alpha b = 0.09985b$, $\rho_f = 2.3769 \times 10^{-3}$ slug/ft^3 (1.23 kg/m^3). The non-dimensional critical speeds for various values of bend-twist coupling parameter, K_s, are shown in Table 1. We observed very good agreements between the current predictions and the predictions given in [9]. As was done in [9], the speeds in the table are scaled with $U_0 = 400$ mph.

4. Results – hydroelastic behavior of composite cantilevered plates

4.1. Problem setup

All the results mentioned from hereon are for a composite cantilevered plate that is 30 cm long, 10 cm wide with 1.2 cm thick. The plate is a canonical approximation to the more complex tapered hydrofoil used in the experiments in the recirculating cavitation channel at the University of Tasmania in Australia (refs. [16,27,28]). The

Table 1
Critical speeds predicted with the high-order beam formulation for different values of bend-twist coupling coefficient K_s/EI_s, for a 20 ft long, 3 ft wide wing in air with $a = -0.34$, $EI_s = 23.6 \times 10^6$ lb.ft^2, $GJ_s = 2.39 \times 10^6$ lb.ft^2, $m_s = 0.746$ slug/ft, $I_s^{E.A.} = 1.943$ slug.ft, $x_\alpha b = 0.09985b$, $\rho_f = 2.3769 \times 10^{-3}$ slug/ft^3. Speeds are scaled with a 400 mph reference speed.

K_s/EI_s	Relative Flutter Speed		Relative Static-Divergence Speed	
	Ref. [9]	Present Study	Ref [9]	Present Study
−0.2	—	—	0.67	0.67
−0.1	2.14	1.92	0.87	0.88
0	1.49	1.41	1.11	1.10
0.1	1.14	1.11	1.43	1.44
0.25	0.75	0.75	—	—

original geometry had a modified NACA 0009 cross-section profile with a 30 cm long span, and has a linearly tapered planform with a root chord of 12 cm, and a tip chord of 6 cm. The tapered composite hydrofoil was made of alternating layers of T700 unidirectional fibers and non-crimp biaxial E-glass fabrics [Glass-(0°/90°)], all in between an outermost layer of a light basket weave E-glass fabric and an inner core layer of glass mat fabric. The laminae arrangement followed the $[(0°/90°)_g,(\theta)_{5c}, (0°/90°)_{2g},(\theta)_{4c}]_s$ order, where the subscript c is for the carbon reinforced layers, the subscript g is for the glass fabric layers, and the subscript s denotes the sequence is symmetric about the hydrofoil mid-plane. For sake of simplicity, we approximated the tapered multi-layered composite hydrofoil as a cantilevered rectangular unidirectional CFRP plate, where the dimensions was selected to match the first two resonance frequencies in vacuum of the original tapered cantilevered hydrofoil. Following the measurements reported in [28], which followed the ASTM D3039 and D3518 standards, the assumed structural properties of the unidirectional CFRP plate are $E_1 = 117.8$ GPa, $E_2 = 13.4$ GPa, $G_{12} = 3.9$ GPa,$\nu_{12} = 0.25$, and solid density 1540 kg/m^3. The material damping is assumed to be 2% for all the modes, which yields $E_1 = 117.8(1 + 0.04i)$ GPa, $E_2 = 13.4(1 + 0.04i)$ GPa, and $G_{12} = 3.9(1 + 0.04i)$ GPa in the frequency domain. Appendix A gives the relations between E_1, E_2, G_{12}, ν_{12} and EI_s, GJ_s, K_s. The center of pressure is assumed to be at the quarter chord, i.e. $e = 1/2 + a$. The center of gravity and the elastic axis are both assumed to be located at the midchord, i.e. $a = 0$ and $x_\alpha = 0$.

4.2. Influence of fiber angle on the effective structural bending, twisting, and bend-twist coupling stiffness parameters

As shown in Fig. 4.1, the plate's effective stiffness parameters are functions of the fiber angle. The bending stiffness, EI_s, is higher for $\theta = 0°$, and decays rapidly with higher values of $|\theta|$. The twisting stiffness, GJ_s, is highest at $\theta \sim \pm 35°$, and decays to its minimum at $\theta = 0°$ and 90°. The bend-twist coupling stiffness, K_s, is zero at $\theta = 0°$ and 90°, and it is antisymmetric about $\theta = 0°$. For $\theta > 0°$ (i.e. when the fibers are aligned towards the leading edge, see Fig. 2.1), K_s is positive as positive bending deformations (upward, in the lift direction) create a restoring torque leading to nose-down twist, while the opposite is true for $\theta < 0°$.

Hence, the plate's hydroelastic behavior under fluid-induced loads will be different with the material anisotropy. We first show the steady-state response of the composite plate in water, followed by comparison of the plate's dynamic response in air and in water. We will show that

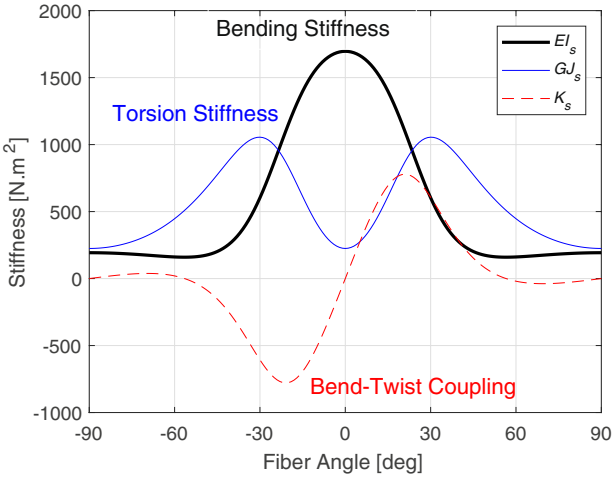


Fig. 4.1. The effect of different fiber angles on effective stiffness components (EI_s, GJ_s, K_s) for the 30 cm long, 10 cm wide with 1.2 cm thick cantilevered composite plate mentioned in Section 4.1, with $E_1 = 117.8$ GPa, $E_2 = 13.4$ GPa, $G_{12} = 3.9$ GPa, $\nu_{12} = 0.25$.

the governing instability mechanism and the critical speeds are very different between the in air and in water cases.

4.3. Steady-state hydroelastic response in water

The steady-state tip bending (h_0^{tip}) and twist-angle (ψ_0^{tip}) deformations, as well as the 3D lift and moment coefficients (denoted as C_{L0} and C_{M0}), are shown in Fig. 4.2 for the composite plate described in Section 4.1, for three different fiber angles, in uniform water flow with speed $U_0 = 15$ m/s at different initial angle of attacks α_0. The moment coefficient and the tip twist deformation ψ_0^{tip} are defined at the beam's elastic axis, and they are positive in the nose-up direction.

As expected, Fig. 4.2 shows that the tip-bending deformations are smallest when the fibers are aligned along the plate's span axis ($\theta = 0°$), where the bending stiffness is the highest, as was shown in Fig. 4.1. The plate undergoes bending-up (positive because of the upward lift) and nose-down (negative) twist, which reduce the effective angle of attack for $\theta > 0°$ because of the material bend-twist coupling; the plate undergoes a bending-up and nose-up (positive) twist that acts to increase the effective angle of attack for $\theta \leq 0°$. Note that hydrodynamic bend-twist coupling caused by the center of pressure being upstream of the E.A. is responsible for the slight nose-up twist for $\theta = 0°$, when the material bend-twist coupling is zero. The lift and moment coefficients are the highest for $\theta < 0°$ and the lowest for $\theta > 0°$, and the difference is more severe at higher speeds. The predicted trends are consistent with experimental and numerical results reported in ref. [16].

In Fig. 4.3, we plot the effect of different flow speeds on the plate tip-deformations and 3D lift and moment coefficients, at an initial angle of attack of $\alpha_0 = 6°$. The static divergence speeds, obtained from solving Eqs. (19) and (20), are shown as vertical dashed lines in Fig. 4.3. For $\theta = 0°$ and $-15°$, static-divergence occurs at about 49 m/s and 29.5 m/s, respectively, where the deformation becomes unbounded. Static divergence occurs earlier for $\theta = -15°$ because of the higher nose-up twist with positive lift that acts to increase the effective angle of attack, which further increases the lift. For $\theta = 15°$, the deformations and loads are bounded even for the highest speed, because of the nose-down twist with positive lift. It is important to note that even well before the static-divergence speed, the deformations can get excessive, as the fluid disturbing force grows with speed square. For example, in Fig. 4.3, for $\theta = -15°$ at $U_0 = 20$ m/s, the pitch deformation is $> 5°$, which, together with the initial geometric angle of attack at 6°, brings the effective angle of attack at the tip to be $> 11°$, and will cause the flow near the tip to stall. Recall that the formulation given in Eqs. (10) and (11) assumed fully attached flow with no separation. We also assume linear elastic material response without failure.

4.4. Dynamic response in water compared to in air

Having discussed the steady-state hydroelastic response of the composite plate in water, here we compare the in water and in air natural frequencies, damping loss factors, critical speeds and governing instability mechanisms. Fig. 4.4 shows the effect of different fiber angles on the predicted natural frequencies for the composite plate mentioned in Section 4.1, both in still air and in still water. By analyzing the mode shapes (only first three modes are shown in Fig. 4.5), we found that the first four modes of the plate in still air and in still water at 0° fiber angle were, respectively: (1st) bending, (1st) twisting, (2nd) bending, and (2nd) twisting. Observe that, for nonzero θ, the mode shapes get convoluted between bending and torsion because of the material bend-twist coupling.

The plots in Fig. 4.4 show that: (1) the first natural frequency has a maximum at $\theta = 0°$ and decreases as θ deviates from 0°, which is consistent with the trend of EI_s vs. θ plot in Fig. 4.1; (2) the second natural frequency, similar to the GJ_s vs. θ plot in Fig. 4.1, has a local minimum at $\theta = 0°$ and $\pm 90°$. The natural frequencies in water are significantly smaller than in air, due to the higher fluid added-mass

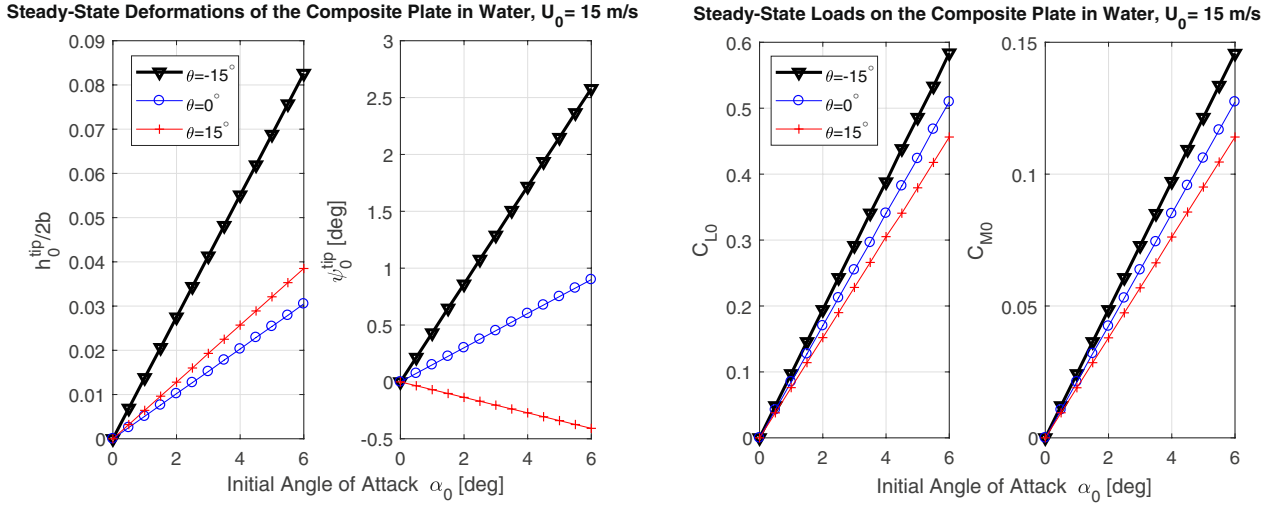


Fig. 4.2. The effect of the fiber angle on the steady-state tip-deformations and 3D lift and moment coefficients at the sectional elastic axis, as defined in Eq. (23), for different angles of attack α_0 for a 30 cm long, 10 cm wide, and 1.2 cm thick, composite cantilever plate in water described in Section 4.1 with uniform inflow with speed U_0 = 15 m/s. The plate elastic properties are E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, and ν_{12} = 0.25.

effect caused by the higher water density. As shown in Eqs. (10) and (11), the added-mass effects depend on the displacement direction [27], and hence the in water to in air frequency ratio varies with mode shape. In still air and in still water, the fluid-induced damping is zero (due to the potential flow model in unbounded domain given in Eqs. (10) and (11)), so the damping loss factors for all the modes are equal to the material damping loss factor, which was assumed to be 0.04.

The series of plots in Fig. 4.6 show the effect of flow speed on the in air natural frequencies and damping loss factors for the composite plate discussed in Section 4.1. For $\theta = 0^\circ$, the natural frequency of the first mode appears to remain nearly constant; for the second mode, however, the frequency decrease with increasing speed, and the first and second mode frequencies converge as speed increases. The damping loss factor, η, of the first two modes initially increases with speed; however, η for mode 2 starts to decrease for $U_0 > \sim 1000$ m/s and eventually crosses zero at U_0 = 1280 m/s, which is the critical flutter speed, and is denoted by the vertical dashed-dotted line in Fig. 4.6. The root-locus plot in Fig. 4.7 shows the variation of the complex eigenvalues (poles) of this vibrating plate in air with different air speeds. Recall that, through Eqs. (8) and (9), we defined the natural frequency as the real component of the complex pole, and the damping loss factor as twice of the ratio of the imaginary to the real values of the pole. Observe in Fig. 4.7 that the imaginary component of the poles of the 2nd mode for $\theta = 0^\circ$, crosses from the positive to negative at the flutter speed at U_0 = 1280 m/s. Since the second mode is torsion, this is called torsion flutter. It is also called coupled-mode flutter, as the flutter mechanism involves convergence of the frequencies of two adjacent modes.

For both $\theta = 0^\circ$ and -15°, we see an emergence of a new pole close to the imaginary axis (with very small real component, i.e. near zero frequency) in Fig. 4.7. The new pole's imaginary component crosses over from positive to negative at the critical static-divergence speed, and is denoted by vertical dashed lines in Fig. 4.6 at U_0 = 1406 m/s for $\theta = 0^\circ$ and U_0 = 825 m/s for $\theta = -15^\circ$. It should be noted that this new pole arises because the circulatory terms (which are associated with memory effects) with $\Omega(k = \omega b/U_0)$ in the fluid damping and stiffness (or disturbing force) terms in Eqs. (10) and (11) yield to non-constant coefficients in the equations of motions, as they depend on U_0 and ω. This phenomenon also appeared in the results of the validation

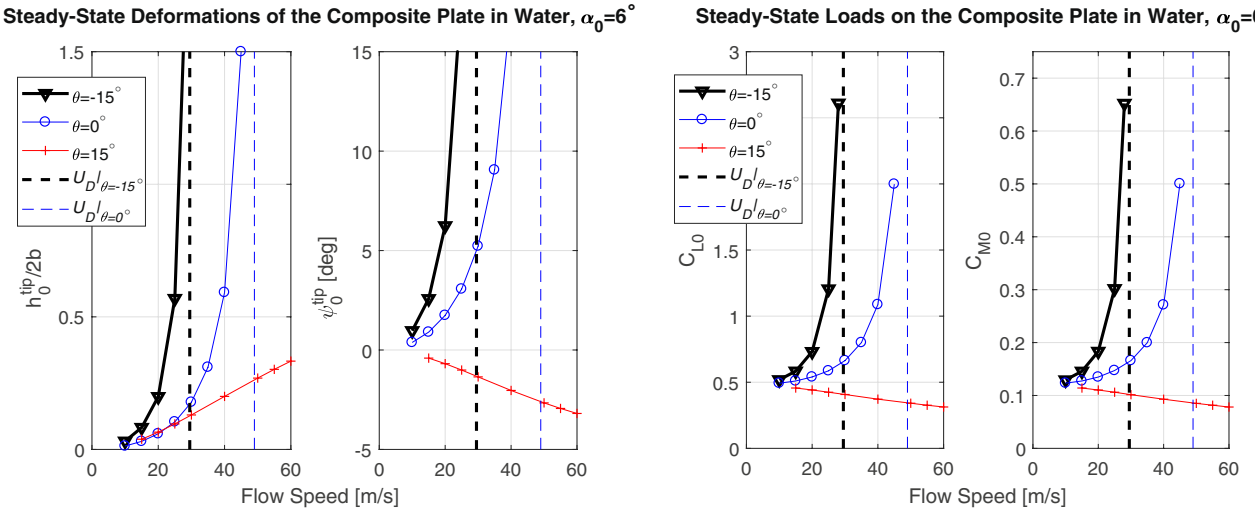


Fig. 4.3. The effect of the fiber angle on the steady-state tip bending and twisting deformations, as well as 3-D lift and moment coefficients, along the elastic axis, as given in Eq. (23), for different water flow speeds U_0 for a 30 cm long, 10 cm wide, and 1.2 cm thick, cantilevered composite plate described in Section 4.1 with an initial angle of attack of 6°. The plate elastic properties are E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, and ν_{12} = 0.25. The vertical dashed lines denote the static-divergence speeds, U_D, obtained by solving Eqs. (19) and (20).

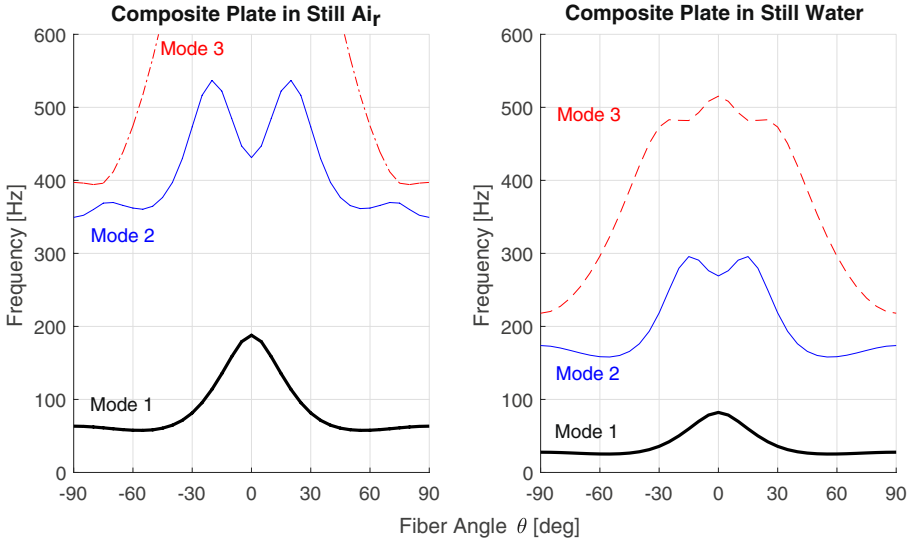


Fig. 4.4. Effect of the fiber angle on the natural frequencies for a 30 cm long, 10 cm wide with 1.2 cm thick, cantilevered composite plate mentioned in Section 4.1 in still air and in still water, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density.

case given in Section 3.1, and was discussed in refs. [23,19]. The fluid damping and stiffness terms, are respectively proportional to speed and speed square, and hence the new mode emerges at high speeds. For $\theta = 15^\circ$, the new mode did not appear for the range of speeds considered.

The 1st mode for $\theta = 15^\circ$ and $\theta = -15^\circ$ involves mixed bending and twisting due to non-zero material bend-twist coupling. Divergence governs for the $\theta = -15^\circ$ case because of the higher loading (and hence fluid de-stiffening force) created by the bend-up and nose-up coupled deformation. Flutter governs for $\theta = 15^\circ$ with the bend-up and nose-down deformations. The damping loss factor for the first mode for $\theta = 15^\circ$ vanished at $U_o = \sim 1017$ m/s.

The plots in Figs. 4.6 and 4.7 show both the natural frequencies and damping loss factors can change with flow speed. In addition, note that the differences between the variation of the natural frequencies and damping loss factors with speed for $\theta = 15^\circ$ and $\theta = -15^\circ$ are solely due to the opposite sign of the material bend-twist coupling and its interaction with the fluid damping and stiffness terms. It is important to note that, for this particular geometry and material, the critical speeds observed for flutter and static-divergence in Figs. 4.6 and 4.7 are higher than the speed of sound, while our model assumes incompressible flow; hence, the results for $U_o > 1/3$ of the speed of sound should only be regarded qualitatively, not quantitatively.

We repeat the same study for the case of our vibrating plate in water, and the results are shown in Figs. 4.8 and 4.9. The natural frequencies are typically much lower and the hydrodynamic loss factors are typically much higher in water than in air at a given flow speed, because the fluid added mass and damping terms are proportional to the fluid density. Similarly, the speed dependency of the natural frequencies is evident at much lower speeds than in air. In addition, note that, since the fluid circulatory terms are proportional to the fluid density, the new mode emerges at a much lower speed in water (Fig. 4.9) than in air case (Fig. 4.7), and that the new mode develops for all the three fiber angles in water. Ref. [29] experimentally and theoretically observed this new mode for isotropic hydrofoils in water. Ref. [29] showed that the critical flutter speed versus solid-to-fluid mass ratio, μ, is not a smooth curve, but rather is a combination of two curves with single-mode (or bending-dominated mode) flutter for $\mu < 0.66$ and coupled-mode (or torsion) flutter for $\mu > 0.66$.

As shown in Figs. 4.8 and 4.9, for the $\theta = 15^\circ$ case, the emergence of

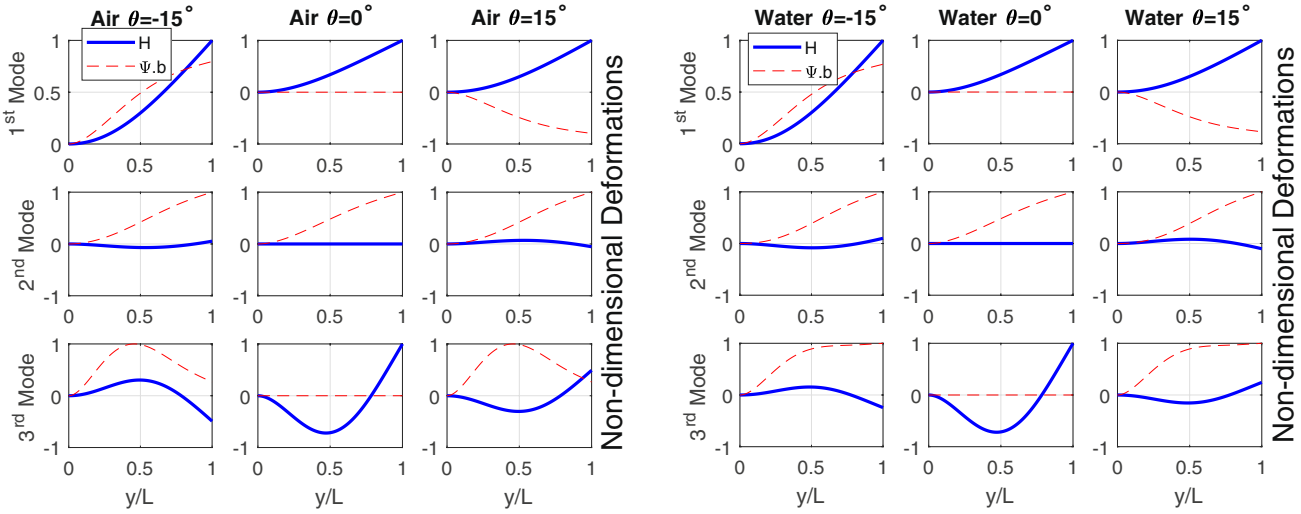


Fig. 4.5. Effect of the fiber angle on the mode shapes of a 30 cm long, 10 cm wide with 1.2 cm thick cantilevered composite plate described in Section 4.1 in still air and water, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density. Thick/solid/blue lines are spanwise bending deformations; thin/dashed/red lines are spanwise twist deformations times half width of the plate.

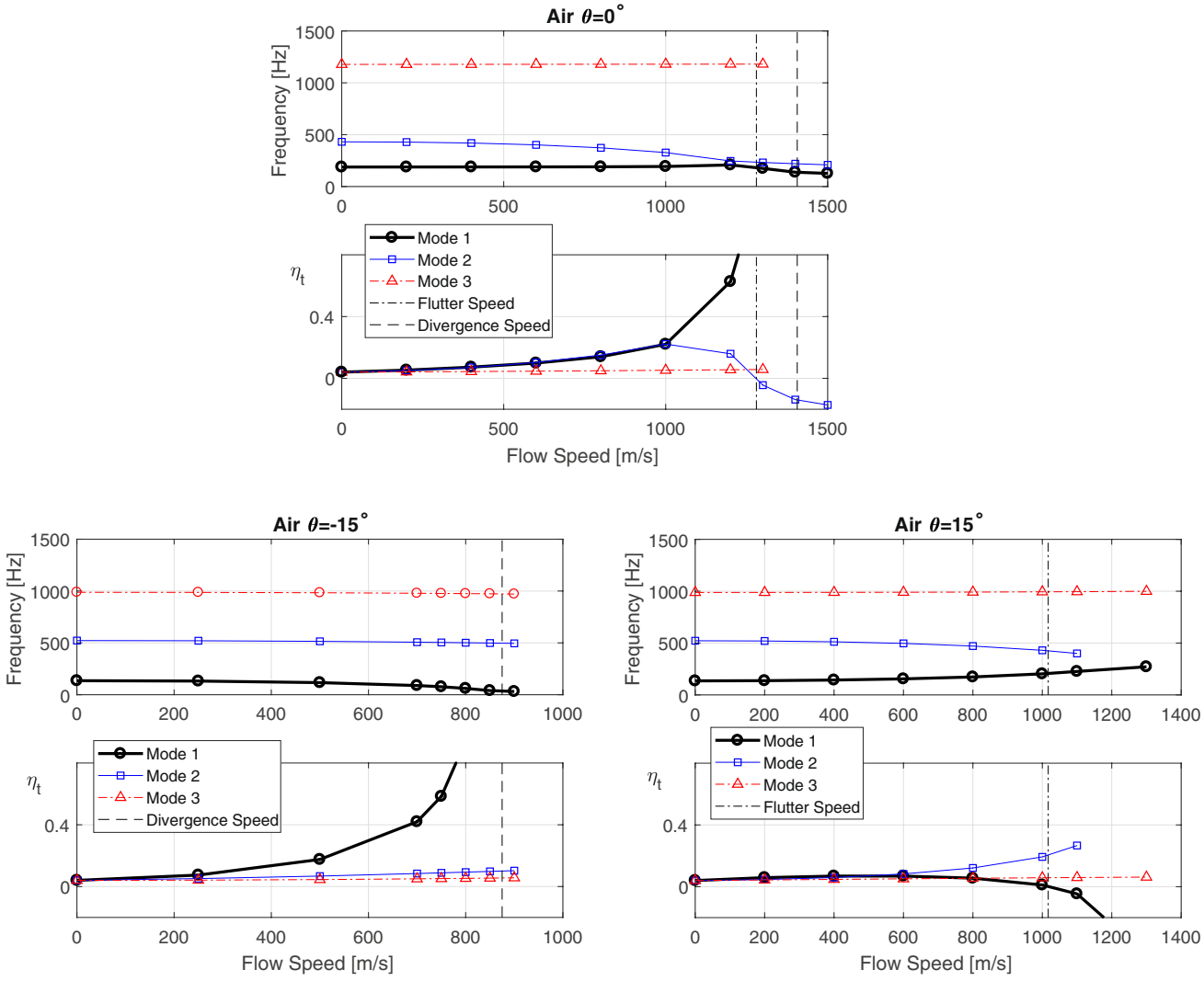


Fig. 4.6. Effect of the flow speed on the natural frequencies and damping loss factors of the cantilevered composite plate in air for three different fiber angles: θ = 0°, ± 15°. The composite plate described in Section 4.1 is 30 cm long, 10 cm wide, 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density.

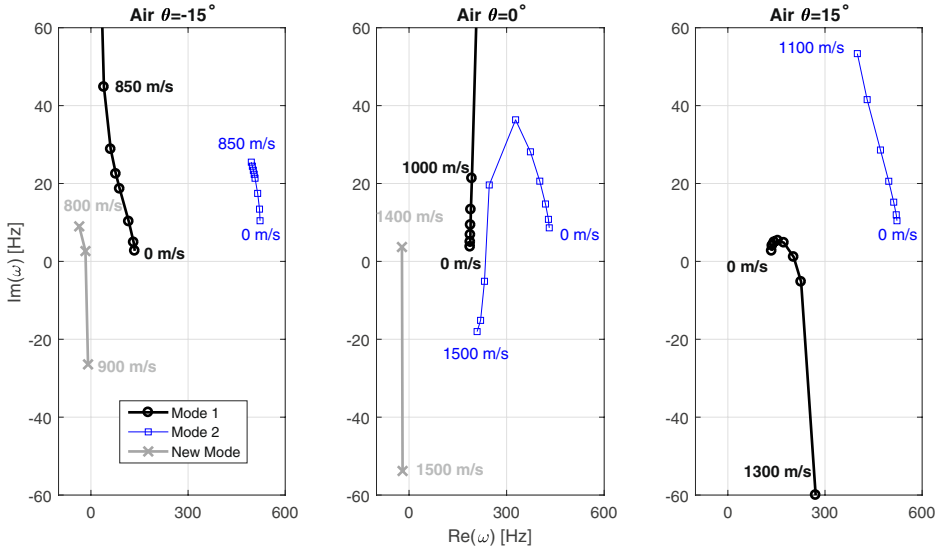


Fig. 4.7. Root-locus plots of the eigenvalues (poles) of a vibrating cantilevered composite plate in air with varying airspeeds for three different fiber angles: θ = 0°, ± 15°. The composite plate described in Section 4.1 is 30 cm long, 10 cm wide, 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density. The Re(ω) is the natural frequency, while damping loss factor is 2.Im(ω)/Re(ω).

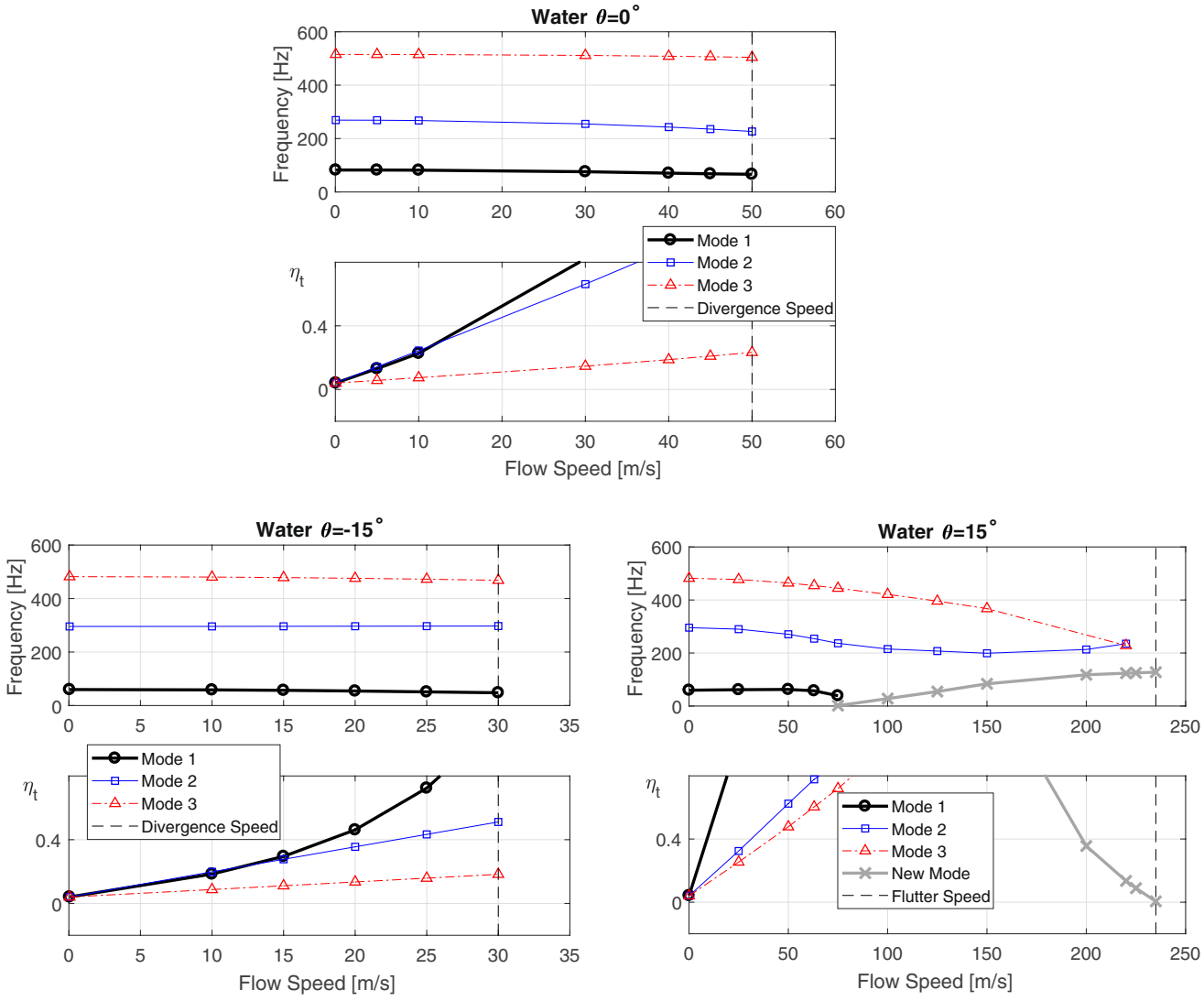


Fig. 4.8. Effect of the flow speed on the natural frequencies and damping loss factors of the composite plate in water for three different fiber angles: θ = 0°, ± 15°. The composite plate described in Section 4.1 is 30 cm long, 10 cm wide, 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density.

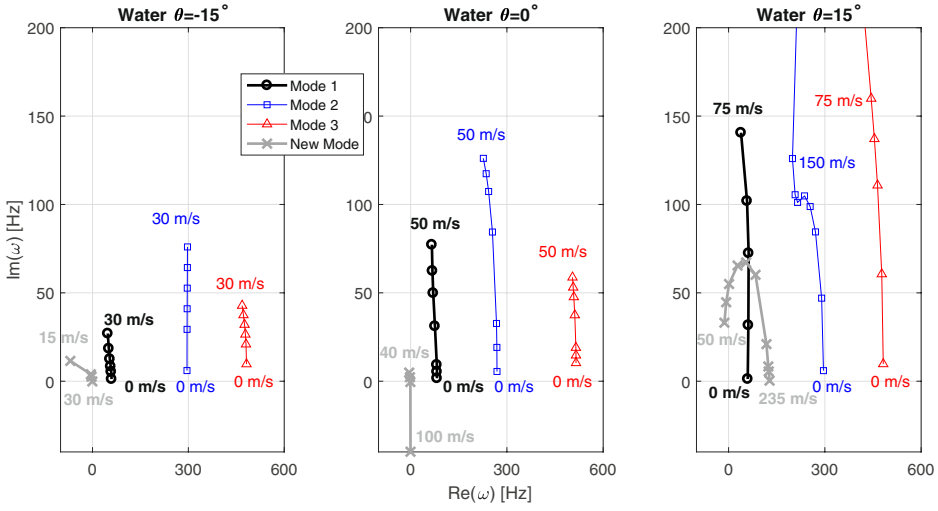


Fig. 4.9. Root-locus plots of the eigenvalues (poles) of a vibrating composite plate in water with varying flow speeds for three different fiber angles: θ = 0°, ± 15°. The cantilevered composite plate descibed in Section 4.1 is 30 cm long, 10 cm wide, 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density. The Re(ω) is the natural frequency, while damping loss factor is 2.Im(ω)/Re(ω).

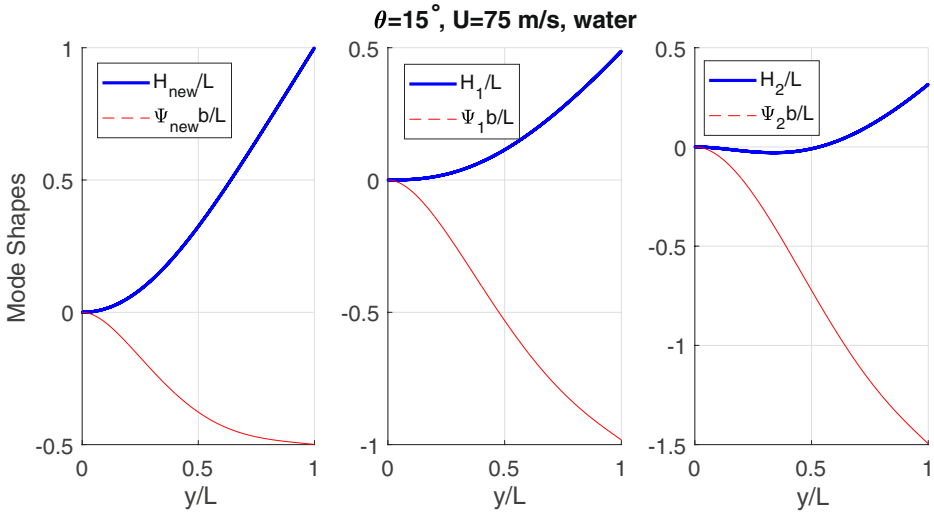


Fig. 4.10. The predicted mode shapes of the new mode and the original first two structural modes for θ = 15° at U_O = 75 m/s in water.

the new mode at U_O = 50 m/s coincided with the rapid decay of the 1st mode frequency. The mode shape of the new mode when it emerges was mostly bending with some twist caused by material and hydrodynamic bend-twist coupling (Fig. 4.10). The new mode first emerged with a negative real value at 50 m/s which increased with speed; this new mode attained a positive vibration frequency for $U_O > 73$ m/s. At U_O = 235 m/s, the damping loss factor of the new mode became zero and the plate entered into a flutter instability. Since the crossover from positive to negative damping is at a positive vibration frequency, some researchers have referred to this instability as dynamic-divergence (e.g. ref. [23]).

For the cases of $\theta = 0^\circ$ and -15° in water, Figs. 4.8 and 4.9 show that the new mode's real value was always slightly less than zero, and its imaginary value eventually crossed-over from being positive to negative at about U_O = 50 m/s and U_O = 30 m/s, respectively. This crossover speed is the static-divergence speed, and is where we observed the plate deformations and loads to grow unboundedly in the steady-state results shown in Fig. 4.3. The mode shape of the new mode was a mixture of bending and twist deformation at the divergence speed (Fig. 4.11). In contrast to the case with $\theta = 15^\circ$, the frequency of the 1st mode did not significantly decay before the emergence of the new mode. It is important to note that static divergence for $\theta = 0^\circ$ and -15° occurred when the frequencies of the structural modes (i.e. modes that are present in quiescent fluid) were still positive. This phenomenon, which was also observed in refs. [19,23], could be dangerous, as static-divergence may occur suddenly without advance warning.

To complement the eigenvalue analyzes, in Fig. 4.12 we show sample time histories of the tip bending and twist of the composite plate in water with $\theta = 15^\circ$, before and after the new mode goes unstable. The plots in Fig. 4.12 are computed by releasing the system at the zero reference time ($t = 0$) from a deformation state where the amplitude of the bending and twisting deformations are perturbed by $1/100^{th}$ of their steady-state values for the stated inflow speeds and for an initial angle of attack of $\alpha_0 = 6^0$. Note that U_O = 73 m/s is the speed where the new mode first attains a positive frequency, but with a very high damping ratio. The frequency of the deformations in the time history plots agree with the predicted frequencies of the new mode shown earlier in Figs. 4.8 and 4.9. The time histories show that the vibrations due to perturbation die out when $U_O < 235$ m/s (pre-flutter), because of positive damping, the amplitude of the vibrations remain constant at the critical flutter speed, and the amplitudes increase with time for post-flutter speeds.

In Figs. 4.13 and 4.14, we plot dynamic to steady-state ratios of the deformations and the aero/hydrodynamic load spectra. We make the following remarks:

1) The peaks in the deformation and load spectra mostly agree with the separately predicted pole locations. Note that the excitation loads (F_z^{exc} and M_y^{exc}) have only real frequencies, so peaks in the frequency response are elevated when the complex valued poles are close to the real-axis (i.e. if they have small damping values). The flutter speeds (e.g. U_f = 1280 m/s at $\theta = 0^\circ$ in air, U_f = 1017 m/s at

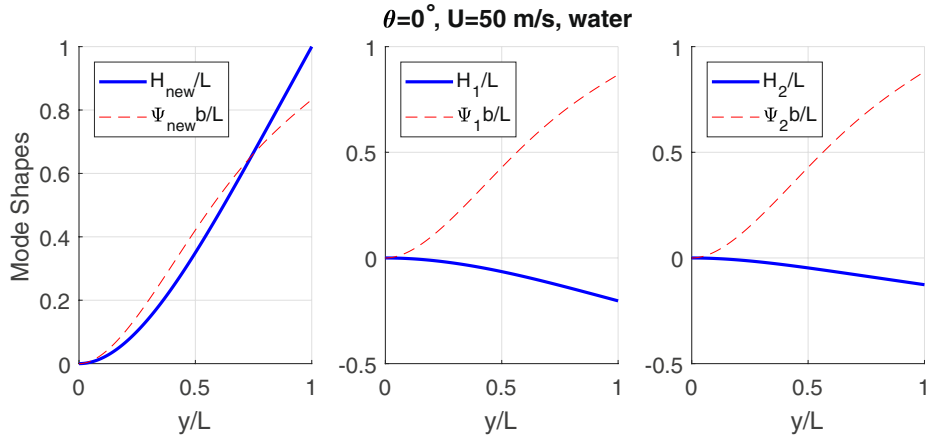


Fig. 4.11. The predicted mode shapes of the new mode and the original first two structural modes for θ = 0° at U_O = 50 m/s in water.

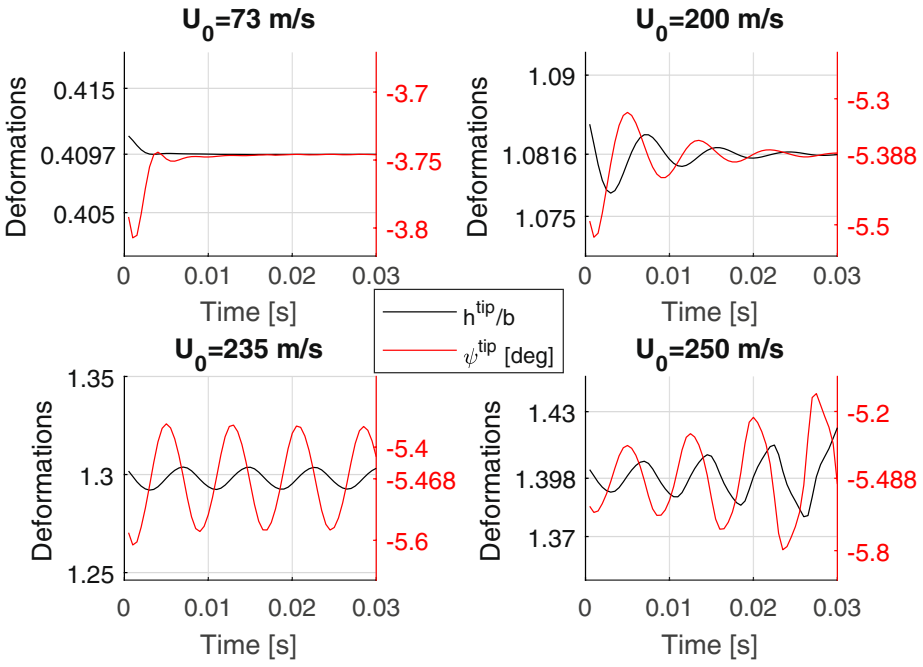


Fig. 4.12. Sample time-response plots of the tip bending and twist angle deformations for the vibrating composite plate described in Section 4.1 at different water speeds. The results are for fiber angle 15°. The cantilevered composite plate is 30 cm long, 10 cm wide, and 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density. The solution assumes 2% material damping for all the vibration modes. The initial conditions are 1/100th of the steady-state values for 6° initial angle of attack at the stated inflow speeds.

θ = 15° in air, and U_f = 235 m/s at θ = 15° in water) are correctly captured; at these speeds, the deformations and loads increased very rapidly;

2) We scale the deformations and loads with their steady-state values, so all the plots start from 1.0 at ω = 0 in Figs. 4.13 and 4.14. However, the unscaled deformation and load spectra in water for the case of the θ = -15° case are separately given in the plots of Fig. 4.15, which show increased steady-state deformations and the presence of the new mode at near zero real frequency, before the static-divergence instability at 29.5 m/s. In Fig. 4.15, the amplitudes of the excitation loads are chosen so that they are equal to the steady-state fluid forces that would been exerted by the fluid if the

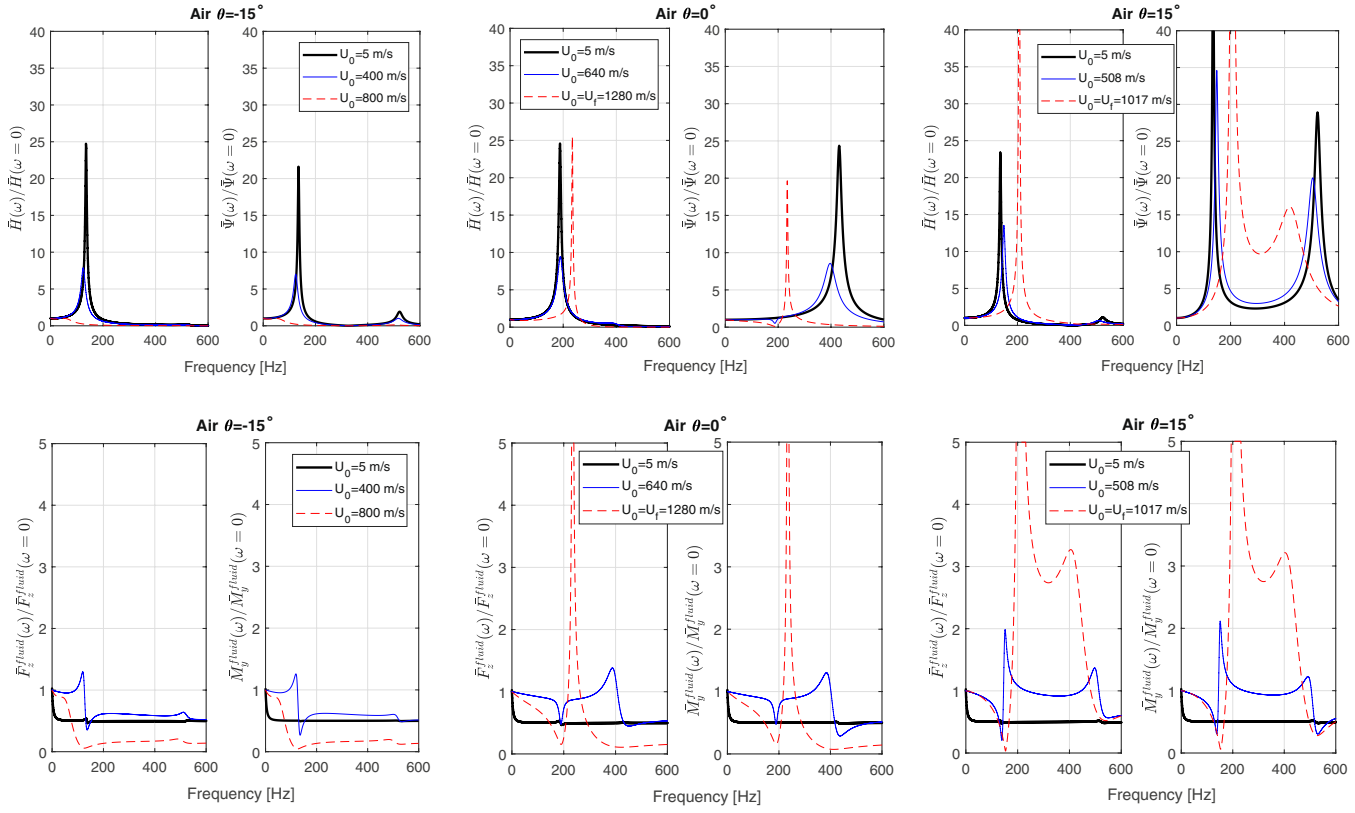


Fig. 4.13. Frequency response spectrums of the deformations and aerodynamic loads on the vibrating cantilevered composite plate at different air speeds. The results are for fiber angles 0° and ± 15°. The deformations and loads are scaled with respect to their steady-state values (i.e. with their values at zero excitation frequency). The composite plate described in Section 4.1 is 30 cm long, 10 cm wide, and 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density.

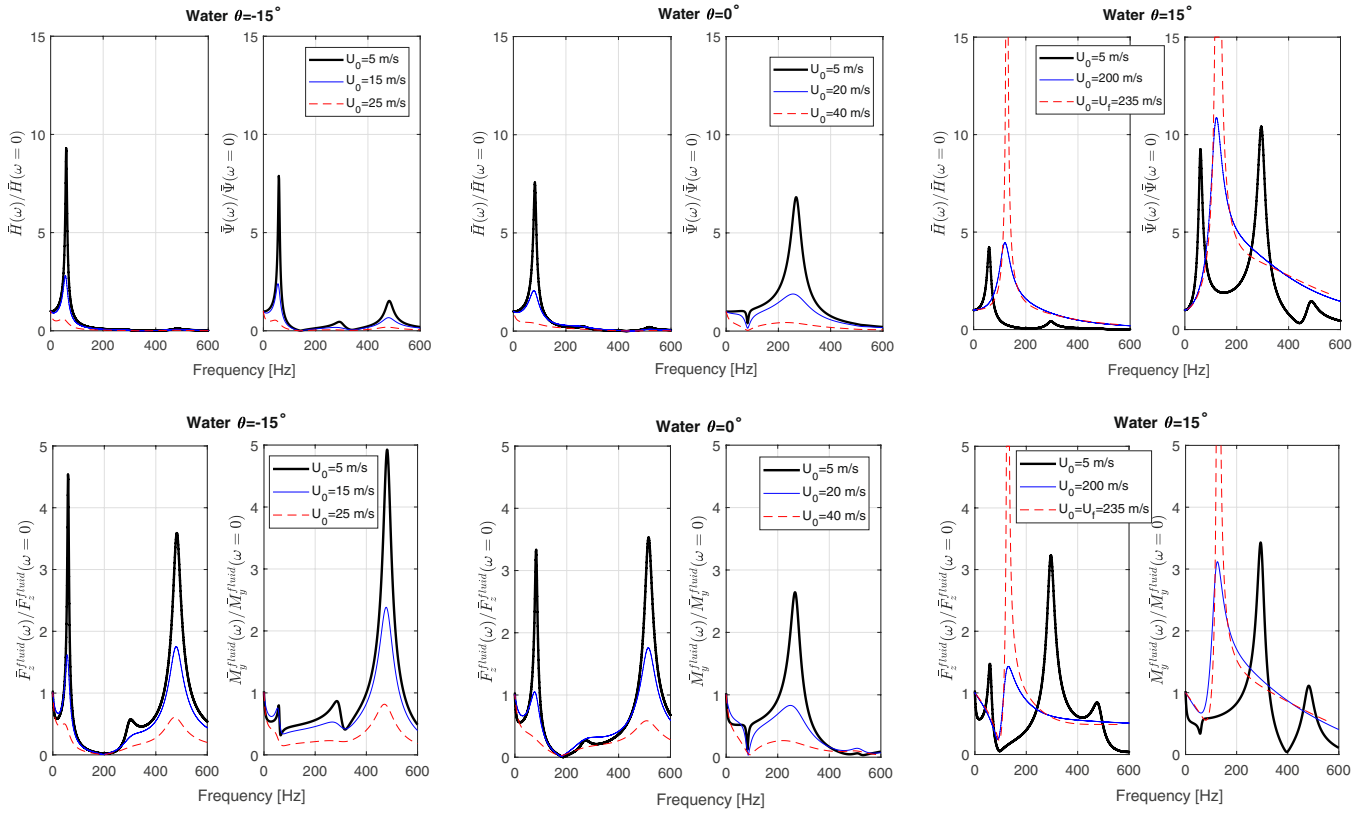


Fig. 4.14. Frequency response spectrums of the deformations and hydrodynamic loads on the vibrating cantilevered composite plate at different water speeds. The results are for fiber angles 0° and ± 15°. The deformations and loads are scaled with respect to their steady-state values (i.e. with their values at zero excitation frequency). The composite plate described in Section 4.1 is 30 cm long, 10 cm wide, and 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density.

initial angle of attack was 6^0;

3) For most of the cases without flutter, the dynamic to steady-state deformation spectra get lower with higher speeds, because the flow-induced damping, in general, increases with flow speed. This effect is more obvious in water at low speeds, because the fluid damping term is proportional to the fluid density;
4) For both the in air and in water cases, even for $\theta = 0°$, the twist spectra include peaks at the frequencies of the bending modes at nonzero flow speeds. This is because the flow-induced bend-twist coupling effects (the de-stiffening effect of the twist motion on the lift force is nonzero, but the de-stiffening effect of bending motion on lift moment is zero because of potential flow assumptions – see Eqs. (10) and (11)); the mixed bend-twist deformation can also be seen in the mode shape plots of Fig. 4.11.

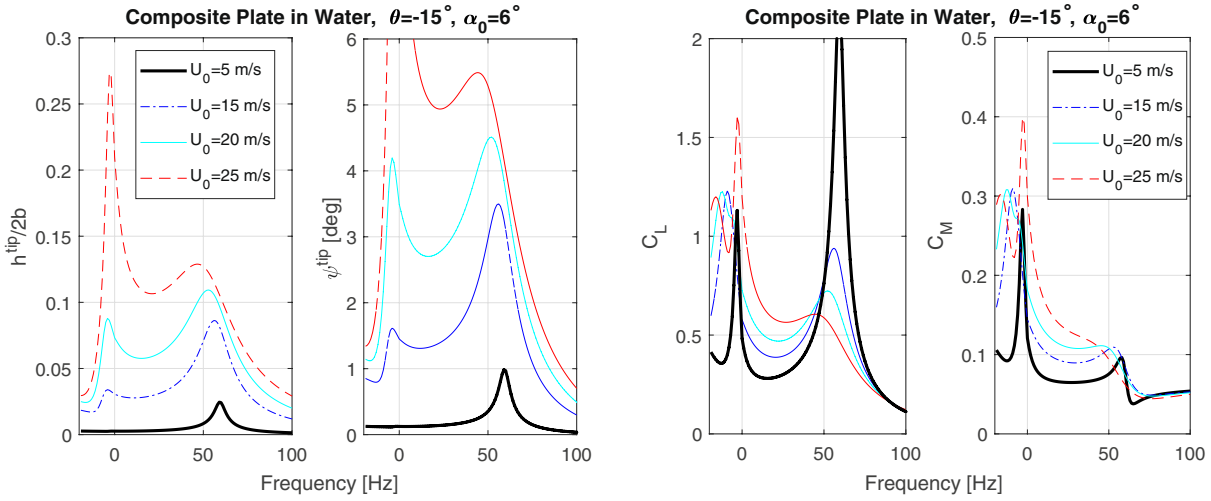


Fig. 4.15. Frequency response spectrums of the deformations and hydrodynamic loads on the vibrating cantilevered composite plate at different water speeds. The results are for fiber angle $-15°$ and for a harmonic excitation force with an amplitude that is equal to the steady-state load when the plate has a 6° angle of attack. Unlike Fig. 4.14, the response is not scaled with the steady-state value and hence the emergence of the static-divergence instability near zero frequency is now visible. The cantilevered composite plate described in Section 4.1 is 30 cm long, 10 cm wide, and 1.2 cm thick, with E_1 = 117.8 GPa, E_2 = 13.4 GPa, G_{12} = 3.9 GPa, ν_{12} = 0.25, and 1540 kg/m^3 mass density.

5) If we scale the fluid force and moment terms with a length scale of b, time scale of $1/\omega$, and mass scale of $\rho_f b^3$(hence the total force is scaled by $\rho_f b^4\omega^2$), we would see a :
[$1 : 1/k : 1/k^2$] proportionality between:
[Fluid added mass loads : Fluid damping loads : Fluid de-stiffening loads],

where $k = b\omega/U_0$ is the reduced frequency. Therefore, within a given fluid medium, the fluid de-stiffening forces dominate the low frequency (k) response and the fluid inertial forces dominate the high frequency response. In Figs. 4.13 and 4.14, the maximum frequency is fixed at $f_{max} = 600$ Hz.

For the case of water in Fig. 4.14, the typical speeds are lower than the typical speeds shown for the case in air in Fig. 4.13. Therefore, between the 0 to 600 Hz frequency range, k values in water are higher than in air, and using the proportionality above, the fluid added mass have a greater influence across most of the prescribed frequency range. Note that fluid added mass loads are proportional to ω^2 times the deformations (Eqs. (24), (25), and (18)); hence, the load spectra in water, which are added mass dominated, at high frequencies are considerably higher than those in air. Such high dynamic loads at high frequencies are dangerous to marine structures, as they can lead to accelerated fatigue. The excessive deformations can also cause vessel stability and control issues, if the lifting surface is used to support the weight of the surface vessel, or as ride control or steering device. Note that excessive flow-induced vibrations have forced ship builders to remove energy saving devices (e.g. asymmetric pre-swirl stators, where the stator blades are effectively cantilevered foils) and prevented ship operators from reaching the top speed of the vessel. Comfort and stability of sensitive instrumentations (e.g. cameras) is another issue with excessive vibrations.

To give a broader picture, we show in Fig. 4.16 the effect of fiber angle on the critical speeds for static-divergence and flutter instabilities for the same cantilevered composite plate introduced in Section 4.1. The critical speeds in water are much lower than in air, because the flow-induced forces, being proportional to the fluid density, are larger in water. In general (except the $\theta = 5^\circ$ case in water), when $\theta > 0$, flutter instability is more prominent; on the other hand, static-divergence instability is more prominent when $\theta < 0$. Again, some of the critical speeds shown in Fig. 4.16 are very high and one may need to include more advanced models (with consideration to flow compressibility, flow separation, cavitation, and material failure) to conduct accurate quantitative studies, if such high-speed operating conditions are of interest. For instance, Fig. 4.16 includes the "stall" speeds, which are lower than static-divergence speeds. Based on the experimental measurements shown in ref. [16], stall occurred on the tapered NACA 0009 hydrofoil when the mean (averaged over the span) effective angle of attack $\alpha_0 + 2\psi^{tip}/\pi$ was 9° or higher. Note that the $2\psi^{tip}/\pi$ factor is tip twist weighted by a sine function representing the first twisting mode shape using the classical beam-twist theory [16]. Moreover, based on the measured failure load of 4.63 kN for the composite hydrofoil with $\theta = 30^\circ$ reported in ref. [16], the failure speed of the hydrofoil in water is about 20 m/s. Similarly, the failure speed is estimated to be about 30 m/s for the $\theta = 0^\circ$ hydrofoil (because of the higher bending stiffness value for $\theta = 0^\circ$ as shown in Fig. 4.1), and somewhere in between for the $\theta = 15^\circ$ hydrofoil. The failure could occur even earlier for the $\theta = -15^\circ$ hydrofoil because of the higher loading induced by the nose up twist. Hence, material failure and stall are likely to occur for all the fiber angle considered before reaching the predicted static divergence or flutter speed in water. Nevertheless, the quantitative results in Fig. 4.16 are only for this particular composite material and geometry. The critical speeds would be lower for larger or thinner hydrofoils. For instance, if the thickness-to-chord ratio of the same cantilevered plate was reduced to 6%, at $\theta = 15^\circ$, flutter would occurred at 85 m/s in water.

The typical flutter type we observed for lightweight composite lifting surfaces in water involved single-mode flutter, where a new mode, which has a shape similar to the first structural mode (i.e. bending dominated with some twist due to material bend-twist coupling) emerges at some high-speeds and its damping value is eventually reduced due to zero; this is in contrast to the conventional torsion or coupled-mode flutter observed for lifting surfaces in air, where the frequencies of two adjacent modes merge with the damping value of the torsional mode going to zero at the critical flutter speed.

5. Summary and conclusions

We have contrasted the hydroelastic behavior of a composite cantilevered plate with varying material anisotropy and flow speed in water and in air. We modeled the plate with higher-order beam equations with material bend-twist coupling effects and warping resistance terms. We modeled the flow-induced forces and moments by extending Theodorsen's method (refs. [21,22]) in two-dimensions (2D) to three-dimensions (3D) by using Yates' method in ref. [20]. The physical and mathematical models were validated with previous measurements of the flutter, and divergence speeds of composite plates in air from refs. [23,9].

We found that both the steady-state and dynamic response could be tailored by selecting the fiber orientation of the composite materials. The major findings include:

1. A new mode emerges in the root locus plots with non-zero forward

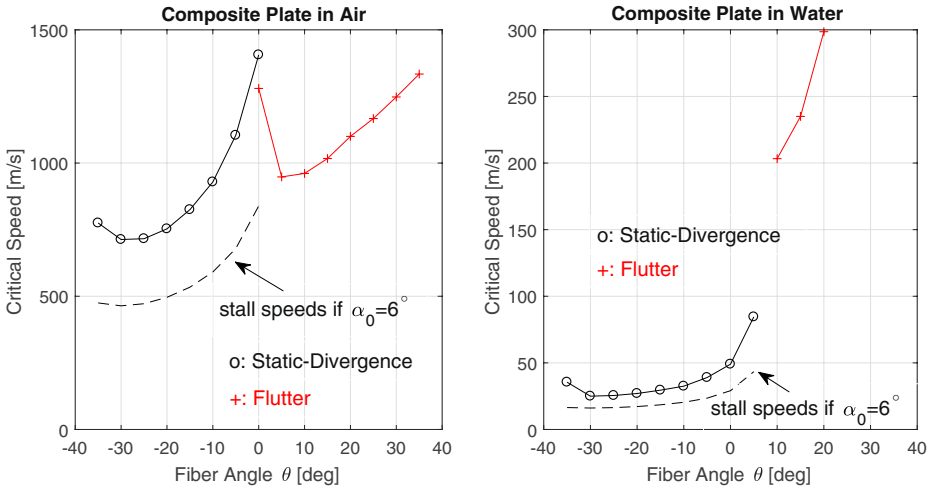


Fig. 4.16. The effect of different fiber angles on critical speeds for flutter and static-divergence instabilities. The stall speeds for the case with $\alpha_0 = 6^\circ$ are also shown, which is when $\alpha_0 + 2\psi^{tip}/\pi$ exceeds 9°. All the results are for a 30 cm long, 10 cm wide with 1.2 cm thick composite cantilevered plate described in Section 4.1 in air and in water, with $E_1 = 117.8$ GPa, $E_2 = 13.4$ GPa, $G_{12} = 3.9$ GPa, $\nu_{12} = 0.25$, and 1540 kg/m^3 mass density.

speed. The new mode arises due to the nonlinear convolution effects exerted by the wake. The new mode emerges only at high speeds, as the circulatory terms in the fluid damping and de-stiffening force terms are proportional to speed and speed square, respectively. Since all the fluid terms are proportional to the fluid density, the new mode emerges at much lower speed in water compared to in air. The vibration frequency of the new mode could be much lower than the fundamental natural frequency of the structural modes in still water. Even at subcritical speeds, the presence of this new mode with low frequency has to be cared for; as shown in the frequency spectrum plots in Fig. 4.14, (for case $\theta = 15°$), the deformations and loads can amplify due to low frequency excitations near this new mode. Note that Ref. [29] experimentally and theoretically observed this new mode for isotropic hydrofoils in water, which was found to be responsible for the observed single-mode flutter. Ref. [29] showed that the critical flutter speed versus solid-to-fluid mass ratio, $\mu = m_s/(\pi\rho_f b^2)$, is not a smooth curve, but rather is a combination of two curves with single-mode (or bending-dominated mode) flutter for $\mu < 0.66$ and coupled-mode (or torsion) flutter for $\mu > 0.66$.

2. Static divergence happens when the fluid disturbing moment is equal to or exceeds the solid elastic restoring moment. For the cases considered here, both in air and in water, static-divergence occurred after a new mode emerged at a certain flow speed with initially very high damping and near zero frequency. At the static-divergence speed, the damping of this new mode vanished, while the frequency of the new mode remained near zero. For this type of static-divergence mechanism, the frequency of the structural bending and torsion modes are nonzero at the static divergence speed. This phenomenon, which was also observed in refs. [19,23], could be very dangerous, as static-divergence may occur suddenly without advance warning.
3. For the composite plates considered here, the flutter mechanism in water is a single-mode mixed bending-twisting type flutter due to the emergence of the new mode with a vanishing damping value at the critical flutter speed. The new mode exhibits a mixed bending-twisting shape with material and/or hydrodynamic bend-twist coupling. In contrast, the typical flutter mechanism in air is a coupled-mode flutter, involving convergence of the frequencies of two adjacent modes, where the damping value of the torsion-dominated mode vanishes at the critical flutter speed. Flutter was only observed for $\theta > 0$, both in air and in water.
4. Due to the higher added-mass effects, the dynamic to steady-state load ratios in water are considerably higher than their values in air, which may bring concerns related to accelerated fatigue.

Future studies could consider adding the missing physics in the current model, such as viscous flow effects with flow separation and stall, cavitation, ventilation, flow compressibility, inelastic material response and material failure. It has been reported in [23,30] that the rigid body modes could be critical and could cause instabilities and should be added to future models. In addition, the influence of sweep, twist, and planform should be investigated.

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Data availability

The raw data required to reproduce these findings are reproducible from solving the mathematical models described in this text. The processed data required to reproduce these findings are available from the plots included in this text. Readers can contact the corresponding author for alternative means of obtaining the data presented here.

Appendix A

For a composite material that is made of a single layer of a relatively weak matrix material with thickness q, which is reinforced with strong and unidirectional fibers, the following equations give the relationship between the equivalent stiffness terms (bending stiffness EI_s, torsion stiffness GJ_s, bend-twist coupling value K_s) and the individual fiber and matrix stiffness terms for a given fiber angle θ. The subscripts 1 and 2 used for material properties E, G, and ν denote to the value of the material property measured in the fiber direction and fiber normal direction within the plane of the lamina, respectively. In the following formulation, θ is the same as was defined in Section 2.1 and Fig. 2.1.

$$EI_s = 2b\left(D_{11} - \frac{D_{12}^2}{D_{22}}\right),\quad GJ_s = 8b\left(D_{66} - \frac{D_{26}^2}{D_{22}}\right),\quad K_s = 4b\left(D_{16} - \frac{D_{12}D_{26}}{D_{22}}\right) \tag{A1}$$

where,

$$\begin{aligned}
D_{11} &= \tfrac{q^3}{12}(Q_{11}\cos^4\theta + 2(Q_{12} + 2Q_{66})\sin^2\theta\cos^2\theta + Q_{22}\sin^4\theta),\\
D_{22} &= \tfrac{q^3}{12}(Q_{11}\sin^4\theta + 2(Q_{12} + 2Q_{66})\sin^2\theta\cos^2\theta + Q_{22}\cos^4\theta),\\
D_{66} &= \tfrac{q^3}{12}((Q_{11} + Q_{22} - 2Q_{12} - 2Q_{66})\sin^2\theta\cos^2\theta + Q_{66}(\sin^4\theta + \cos^4\theta)),\\
D_{12} &= \tfrac{q^3}{12}((Q_{11} + Q_{22} - 4Q_{66})\sin^2\theta\cos^2\theta + Q_{12}(\sin^4\theta + \cos^4\theta)),\\
D_{16} &= \tfrac{q^3}{12}((Q_{11} - Q_{12} - 2Q_{66})\sin\theta\cos^3\theta + (Q_{12} - Q_{22} + 2Q_{66})\sin^3\theta\cos\theta),\\
D_{26} &= \tfrac{q^3}{12}((Q_{11} - Q_{12} - 2Q_{66})\sin^3\theta\cos\theta + (Q_{12} - Q_{22} + 2Q_{66})\sin\theta\cos^3\theta),
\end{aligned} \tag{A2}$$

with,

$$Q_{11} = \frac{E_1}{1 - \nu_{12}\nu_{21}},\quad Q_{12} = \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}},\quad Q_{22} = \frac{E_2}{1 - \nu_{12}\nu_{21}},\quad Q_{66} = G_{12}$$

Note that this formulation is for a single lamina and for materials with multiple lamina these formulas can be generalized as was shown in ref. [6,17].

Appendix B

Fig. A1 and Table A1

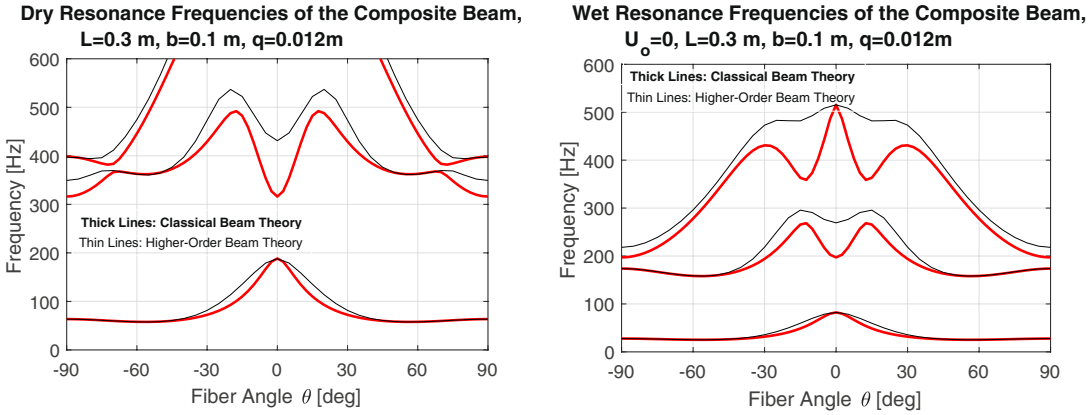


Fig. A1. The effect of choosing between the classical beam theory with bending and twisting degrees of freedom and a higher-order beam theory including warping stiffness effects, on predicting the natural frequencies of a 30 cm long, 10 cm wide with 1.2 cm thick composite cantilevered plate in air and in water, with $E_1 = 117.8$ GPa, $E_2 = 13.4$ GPa, $G_{12} = 3.9$ GPa, $\nu_{12} = 0.25$, and 1540 kg/m^3 mass density.

Table A1
Effect of plate aspect ratio ($L/2b$) on the ratio between the divergence speed predicted by the beam model with warping (U_D) and without warping (U_{D0}), for the case of the composite plate described in Section 4 in water with $\theta = 0^\circ$. In the absence of warping stiffness, this ratio should be 1.

$L/2b$	U_D/U_{D0}
1	2.69
3	1.48
6	1.23

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compstruct.2019.111240.

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